CBSE Test Paper 03

Chapter 12 Thermodynamics

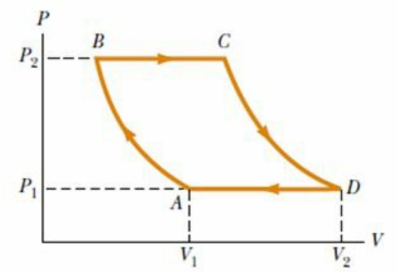
1. A geyser heats water flowing at the rate of 3.0 liters per minute from 27 $^\circ$C to 77 $^\circ$C. If the geyser operates on a gas burner, what is the rate of consumption of the fuel if its heat of combustion is 4.0 $\times$ 10^4 J/g? **1**

   a. 16 g per min
   b. 18 g per min
   c. 14 g per min
   d. 12 g per min

2. If ΔQ is the energy supplied to the system ΔU the change in internal energy, and ΔW the work done on the environment, First Law of Thermodynamics states that. **1**

   a. ΔU + ΔW = 0
   b. ΔQ = ΔU - ΔW
   c. ΔQ = ΔU + ΔW
   d. ΔQ = ΔU $\times$ ΔW

3. An ideal gas is carried through a thermodynamic cycle consisting of two isobaric and two isothermal processes, as shown in Figure. network done in the entire cycle is given by the equation. **1**



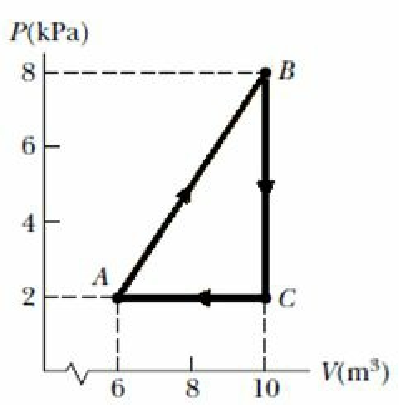
   a. $W_{net} = P_1 (V_2 + V_1) \ln \frac{P_2}{P_1}$
   b. $W_{net} = RP_1 (V_2 - V_1) \ln \frac{P_2}{P_1}$
   c. $W_{net} = P_1 (V_2 - V_1) \ln \frac{P_2}{P_1}$
   d. $W_{net} = P_1 (V_2 - V_1)$

4. A thermodynamic system undergoes a process in which its internal energy decreases

by 500 J. If, at the same time, 220 J of work is done on the system, what is the energy transferred to or from it by heat? **1**

a. Q = -700 J
b. Q = -740 J
c. Q = -730 J
d. Q = -720 J

5. Consider the cyclic process depicted in Figure if ΔE_{int} is negative for the process CA, what are the signs of Q and W are associated with CA? **1**



a. -, 0
b. -, -
c. -, +
d. +, 0

6. Which has a higher specific heat; water or sand? **1**

7. Animals in the forest find shelter from cold in holes in the snow. Why? **1**

8. Show the variation of specific heat at constant pressure with temperature. **1**

9. A steam engine intake steam at 200°C and after doing work exhausts it directly in the air at 100°C. Calculate the percentage of heat used for doing work. Assume the engine to be an ideal engine. **2**

10. What is an adiabatic process? Also give essential conditions for an adiabatic process to take place. **2**

11. State the Second law of thermodynamics and write 2 applications of it? **2**

12. The ends of the two rods of different materials with their thermal conductivities, radii of cross-section and lengths, in the ratio 1:2 are maintained at the same temperature difference. If the rate of flow of heat through the larger rod is 4 cal/s, what is the rate of flow through the shorter rod? **3**

13. Two rods A and B are of equal length. Each rod has its ends at temperatures T_1 and T_2. What is the condition that will ensure equal rates of flow of heat through the rods A and B? **3**

14. An ideal refrigerator runs between -23°C and 27°C. **3**

    i. Find the heat rejected to atmosphere for every joule of work input.
    ii. Also, find heat extracted from cold body.
    iii. Find coefficient of performance of the refrigerator.

15. Derive an expression for the work done during isothermal expansion. **5**